



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Mid Exam Summer 2023
CSE 4611/CSI 411: Compiler Design/Compiler
Total Marks: 30 Duration: 1 hour 45 Minutes

Answer all questions. Figures in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

- 1) a) Show the corresponding inputs and outputs from Lexical analyzer phase to Code Optimizer of compiler based on the following expression where all the variables (P, X, Y, Z, W) are integer type. [3.5+1.5+2]
- $$P = (Y^{(X*Z)} + W) / 21 * 6$$
- b) Is lexical analyzer an essential part of compiler? Explain your reasoning.
- c) Just-in-time (JIT) compilation and ahead-of-time (AOT) compilation are two distinct approaches to code translation and execution. As you are currently enrolled in a compiler design course and have been taught the concepts of compilers and interpreters, please elaborate on the differences between JIT and AOT based on the knowledge you have acquired in class.
- 2) Write down the three address representation of the expression following expression. [3.5]
- $$((a^b)^p)^{(c-d-k)} / (e*j*t) / (g^{(h^2)}) + i - j$$
- 3) Draw syntax tree for the following expression. [3.5]
- $$b * ((a^3)^{2^{(1*3)}})^{2^6} - (b/p/r^3) * a^{2^d}$$
- 4) Prove that the following expression $((b/a^b)^a)$ can be represented by the grammar [3.5+2] given below and then draw the parse tree.
- $$E \rightarrow (ED) | (DE) | BE | C | \epsilon$$
- $$C \rightarrow DE | B | a | b$$
- $$B \rightarrow *BC | ^BC | C | \epsilon$$
- $$D \rightarrow (E) | C | a | b | \epsilon$$

5) Convert the following infix expression to a postfix expression. [3.5]
 $((5 * x) * 3^x) * (x^4 y)^x y / (z + 3/4 * 1 * z) * (2 + x/z)$

6) Convert the following infix expression to a prefix expression. [3.5]
 $b * ((a + 2^{2/3}) - 3 / (b^2))^{((a^b + b) * 3^a c) / ((b^b (b + 1) + c)^2)}$

7) Evaluate the following prefix expression. [3.5]
 $+ * 3 4 - / + 1^2 22 * 3 2 + 1 / 1^2 8 + 2 / 3^4 2$